

HACK THE... HEMISPHERE!!!

shadytel

THE PHONE COMPANY

Karl Koscher (@supersat)
Andrew Green



Tweets893

Following41

Followers1,505

Likes29

Following

Shadytel

@shadytel Follows you

Some people play Fantasy Football. We play Fantasy Phone Company. RT & Follow may imply endorsement. May not be legal in all jurisdictions. Use as directed.

Shady Cove, OR, USA

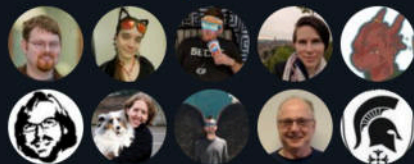
shady.tel

Joined June 2010

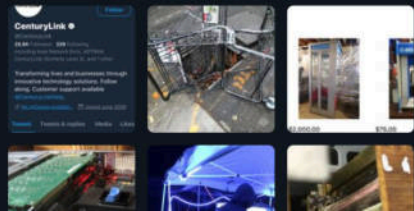
Tweet to

Message

118 Followers you know



47 Photos and videos



Tweets

Tweets & replies

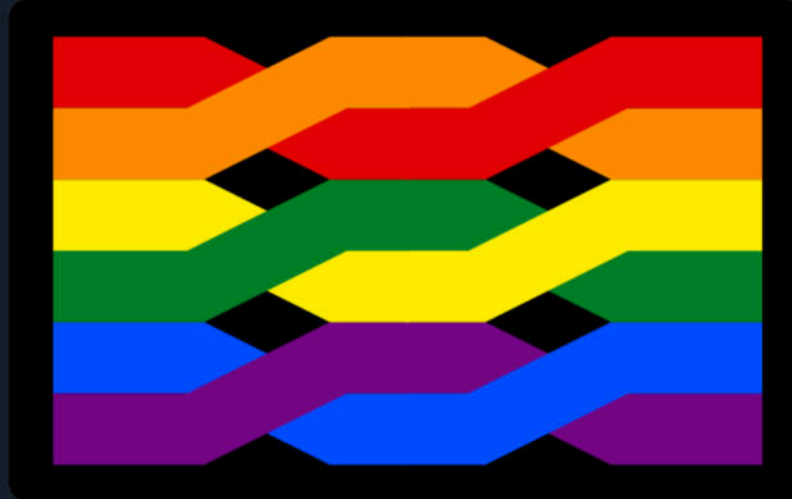
Media

Pinned Tweet



Shadytel @shadytel · 28 Jun 2015

#pride2015 #billingexcellence #twistedpride



3

32

87



Shadytel @shadytel · Mar 31

Celebrate this Trans Day of Visibility with us! We're giving a 10% discount to everyone whose gender is null in the database. Also we're giving one lucky nonbinary customer free line rental this month: Daves Bread! Congratulations, Daves!

11

40

Who to follow · Refresh · View all



#abolish(pol)ICE @banasi...

Follow



Matthew Garrett @mjpg59

Follow



franklin @TheDevilsVoice

Follow



Find people you know

Import your contacts from Gmail

Connect other address books

Worldwide trends · Change

Giants

316K Tweets

#NFLDraft

305K Tweets

Haskins

137K Tweets

#TaylorSwiftTONIGHT

83.8K Tweets

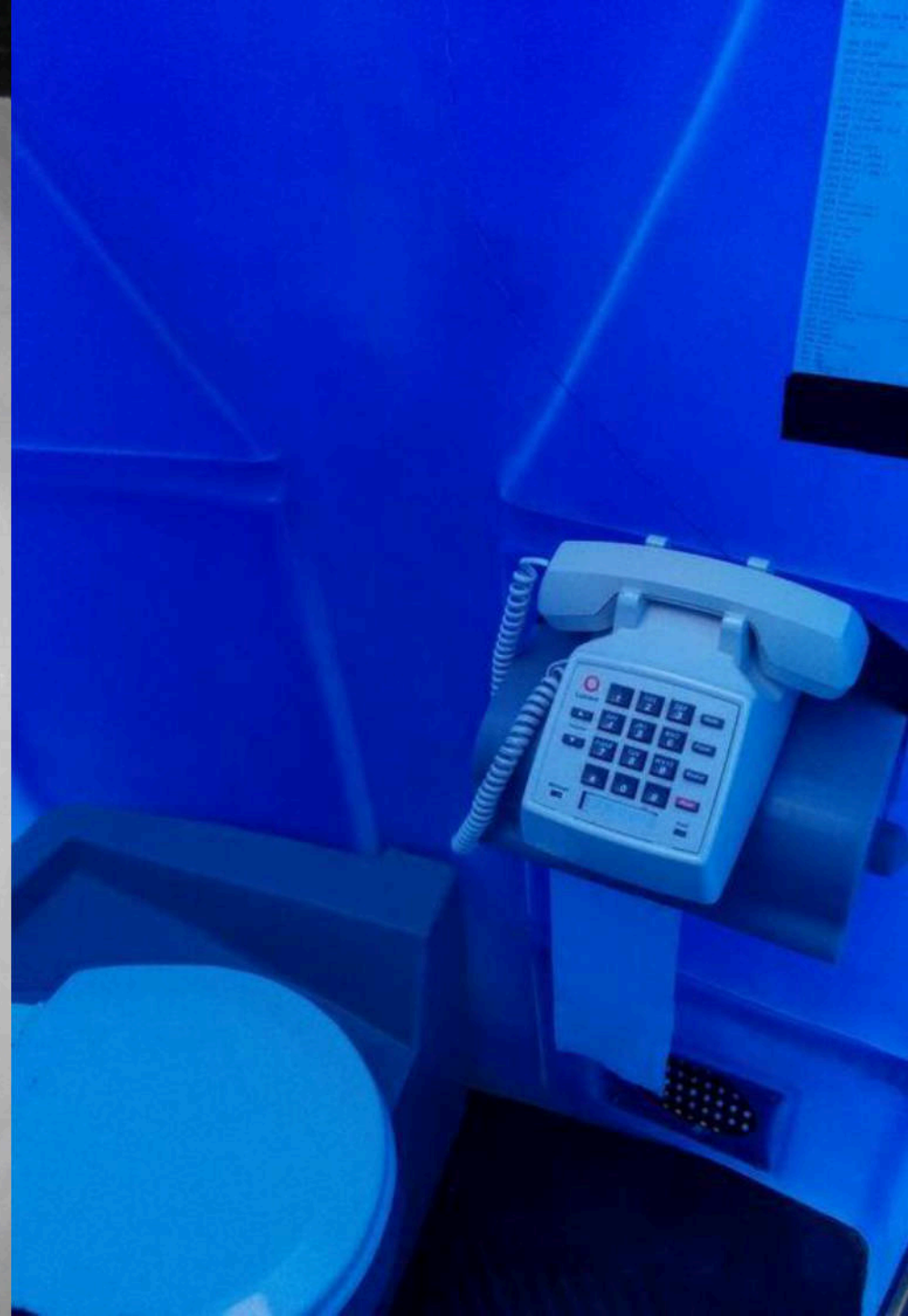
Raiders

119K Tweets



39
ShadyTel Phone Directory
As of July 11 at 1am

3002 CD ISON
3101 Staff
3099 Tree ShadyCamp
3333 the LOL
3210 Duncan's Hamock
3102 Staff Camp
3216 RV Franklin Hu
3999 Annc cct
3133 T Prophet
3995 Cabins DSI Drop
3825 Nick F
3826 Milliways
3003 Black Lodge 1
3004 Black Lodge 2
3005 Garret (Ada's)
3232 Ada's
3323 Alex
3007 PDX
3008 Noisebridge 1
3009 Noisebridge 2
3012 Hak5
3011 Leviathan
3217 RF pal
3323 Alex
3021 Cabin 9
3022 Hack Casual
3023 Registration
3667 DoorPhone
3013 NinjaTel 1
3014 NinjaTel 2
3024 Deprogram
3015 Anything
3016 Nick Lewis (Nerd Creation Lab)
3411 TC Info Desk
3337 Caesar
3020 Nes9

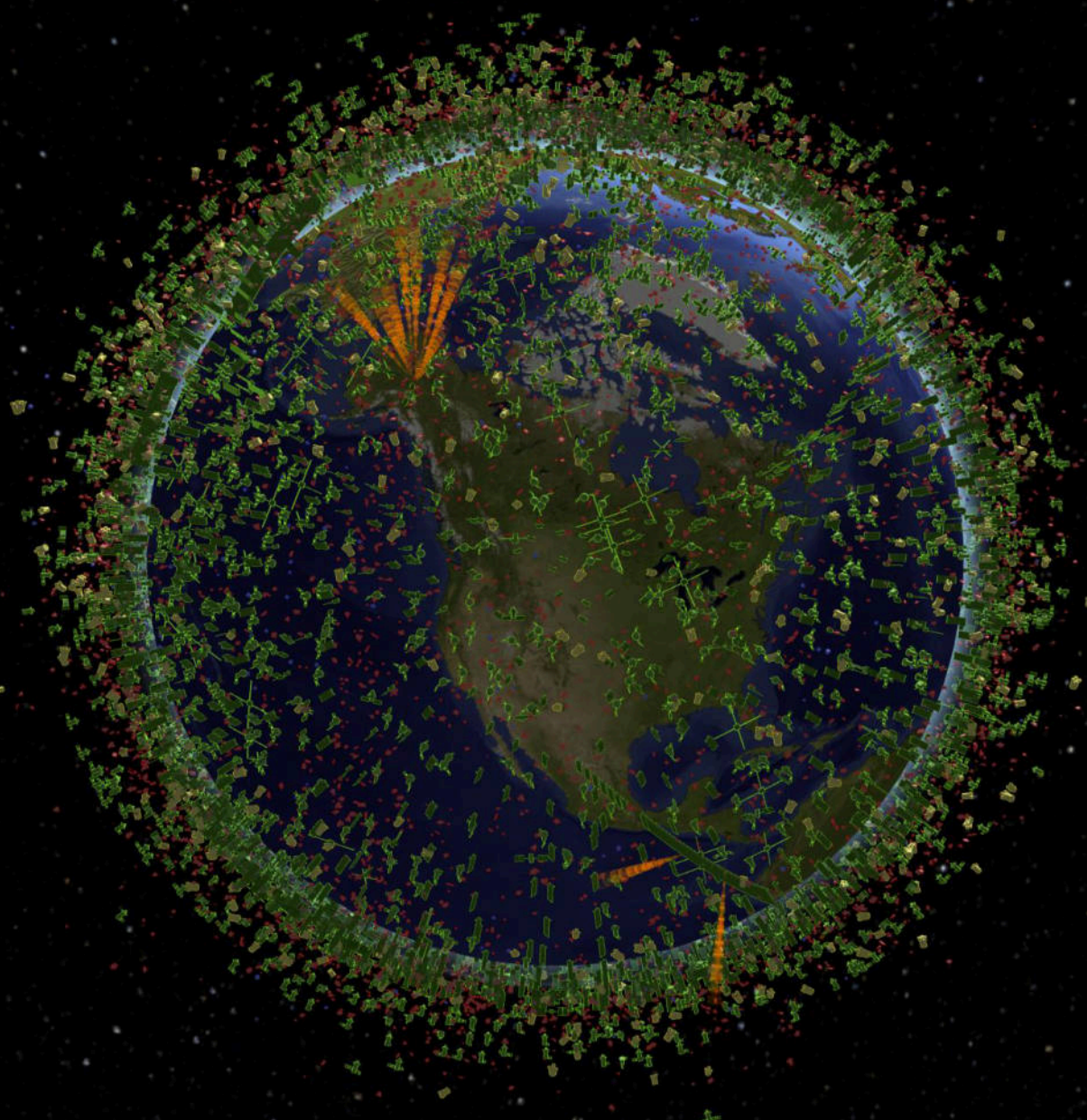


Show Menu

⚠ Special events are not shown

🔗 Copy link to share

Object Type	
■	Payload
■	Rocket Body
■	Debris
■	Unknown



LEO/MEO Satellite Deficiencies

- **Broadcast dishes are aimed at a fix point in the sky**
 - Greatly simplifies installation, maintenance, lifespan, etc.
- **LEO/MEO satellites require *constellations* for continuous coverage**
 - GPS requires 24
 - Starlink requires THOUSANDS





Shadytel @shadytel · Oct 1, 2021

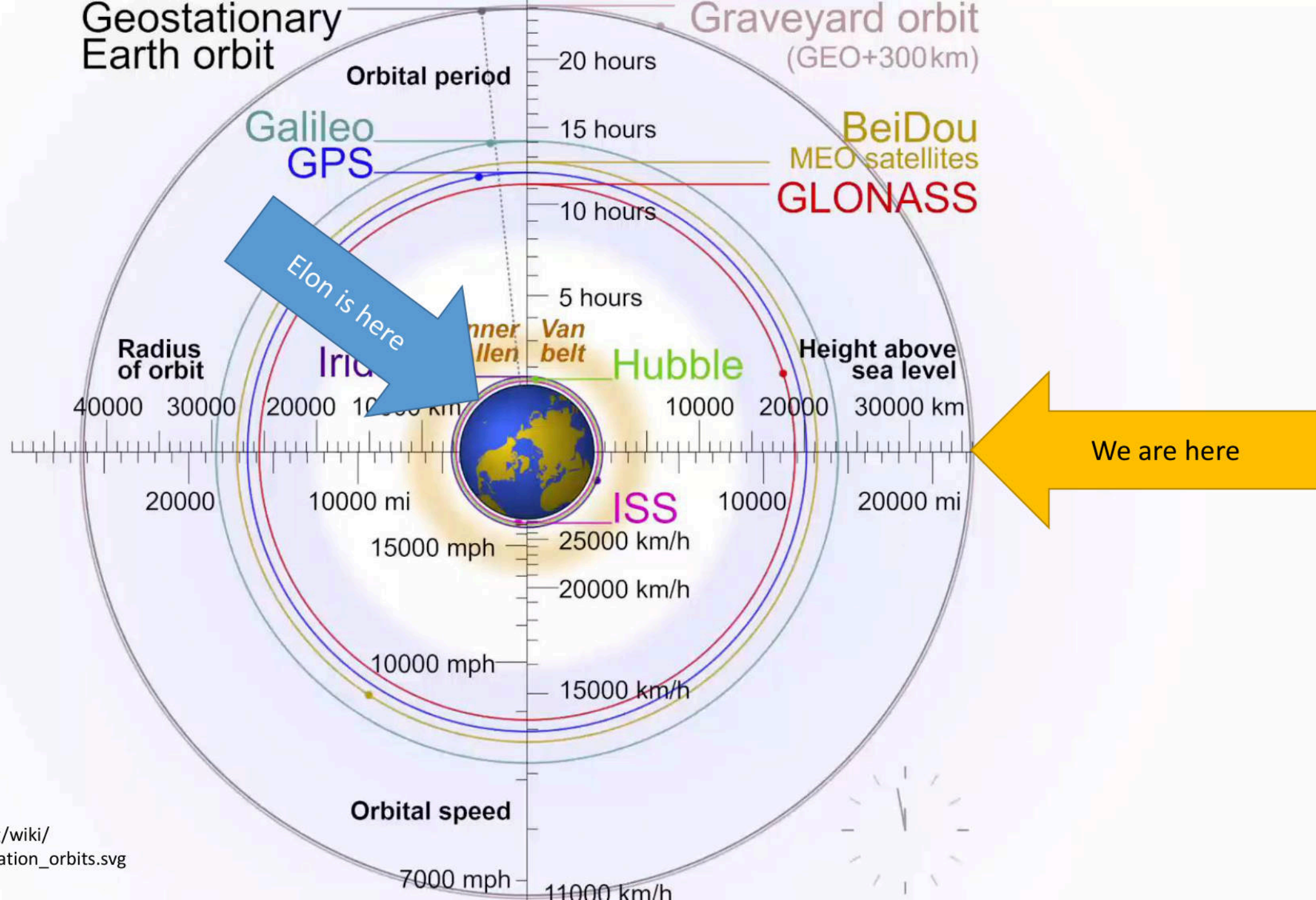


LEO isn't cool.

You know what's cool?

GEO.





Satellite Lifecycles

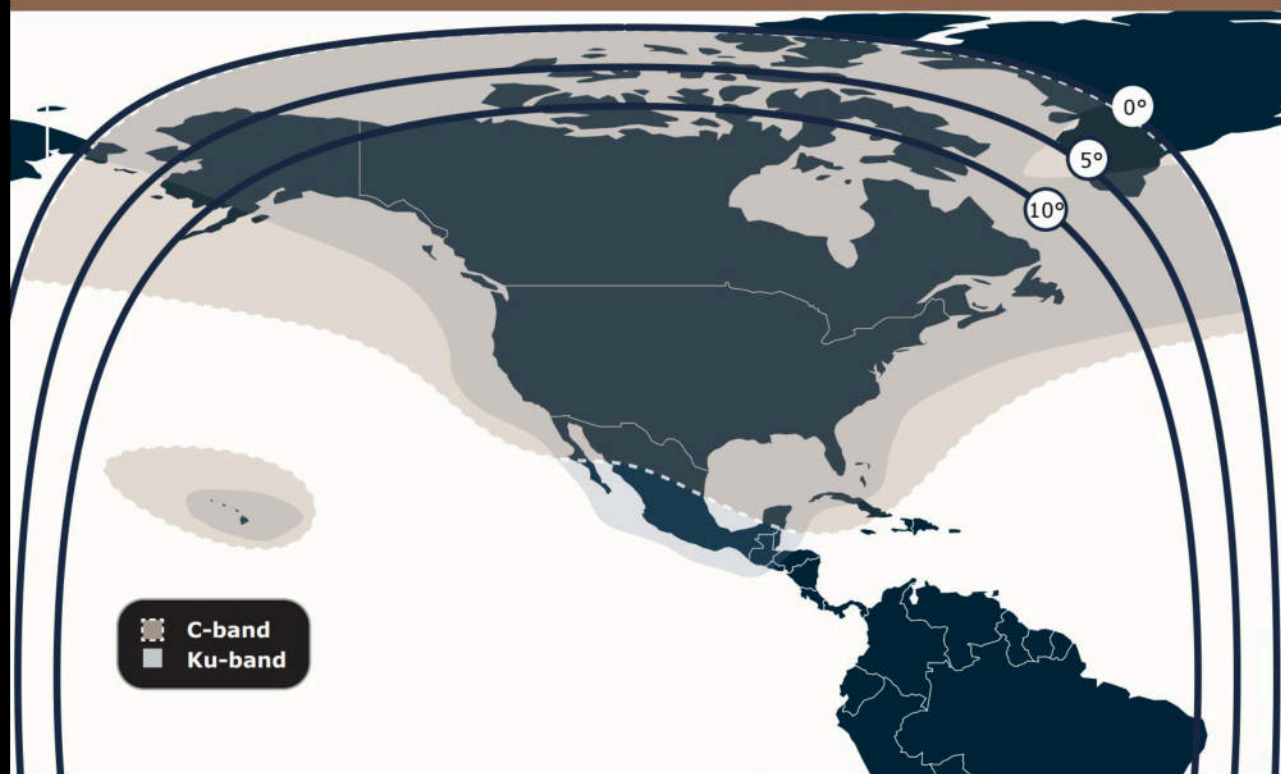
- **Satellites have limited lifespans**
 - Limited fuel for fine-tuning orbit/position
 - Batteries wear out
 - Space debris
 - Radiation
- **Satellites are engineered for a specified lifespan**
- **Geosynchronous satellites moved to “graveyard orbit” at EOL**
 - Minimize risk of space debris
 - Requires far less fuel than deorbiting

An Unprecedented Opportunity...



Anik F1R

Strong North American coverage
in C- and Ku-band frequencies



Key Facts

- Ideal for video distribution to cable systems and for broadcast contribution
- Coverage of all 50 United States and Canada including the Far North
- Flexibility to support virtually any application

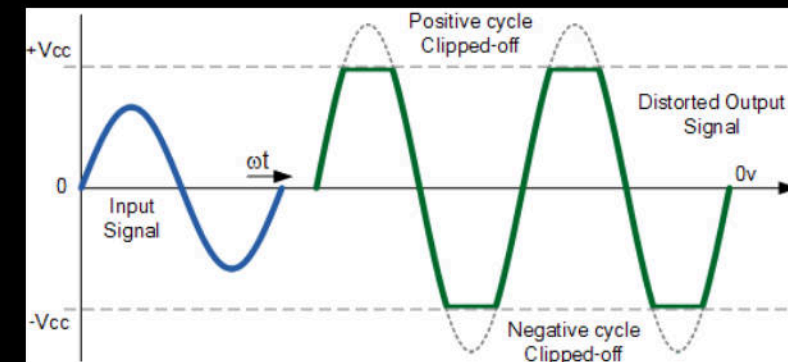
SATELLITE MANUFACTURER
Airbus Defence and Space

TRANSPONDERS
24 C-band @ 36MHz
32 Ku-band @ 27MHz

IN SERVICE DATE
October 2005

Satellite Transponders

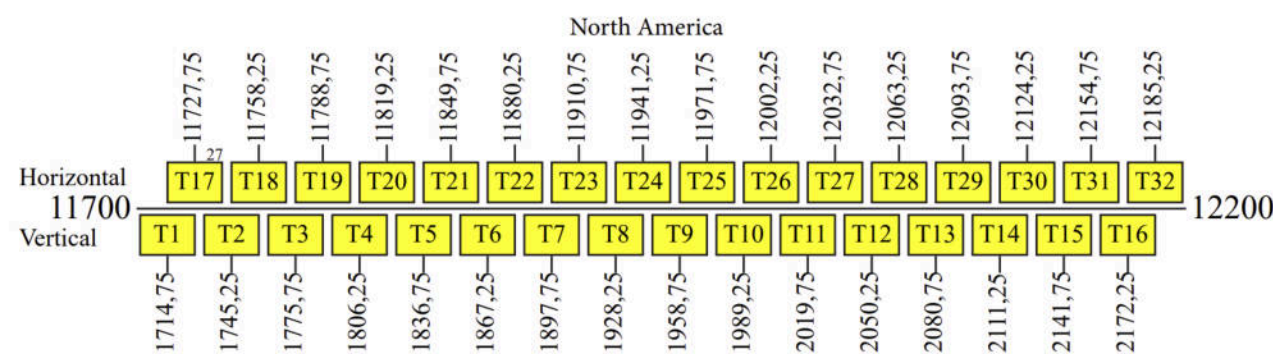
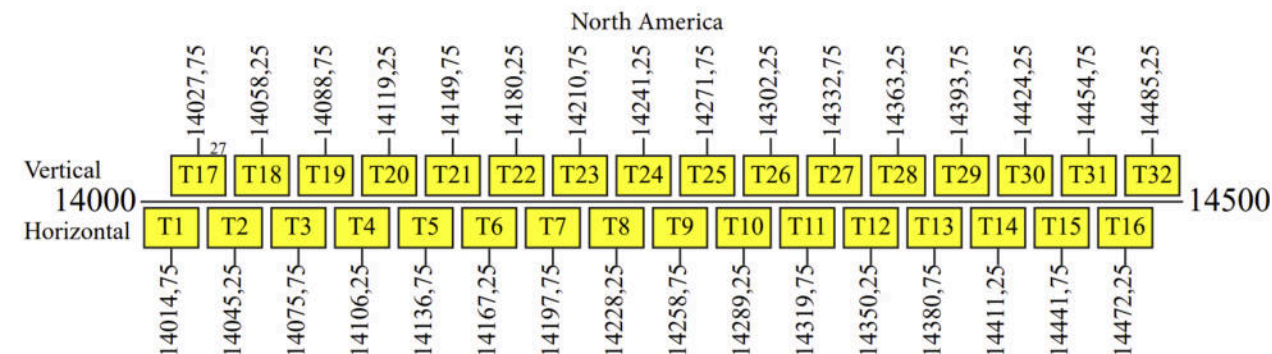
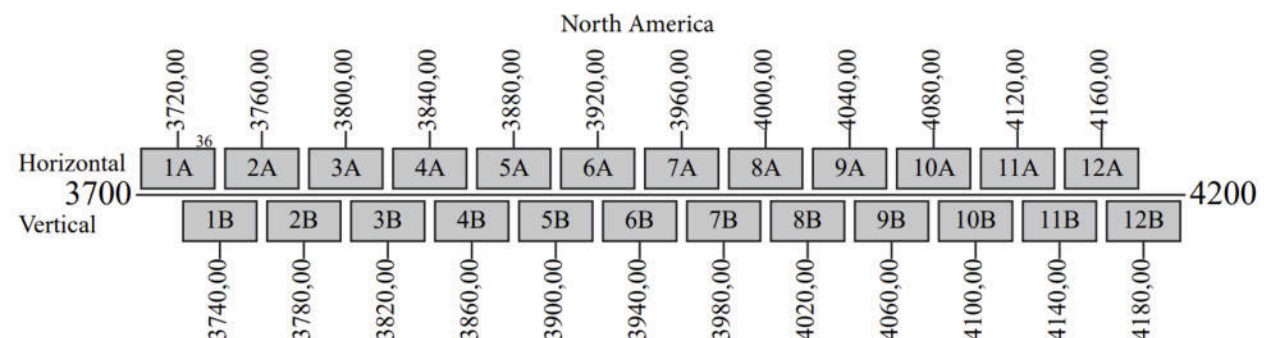
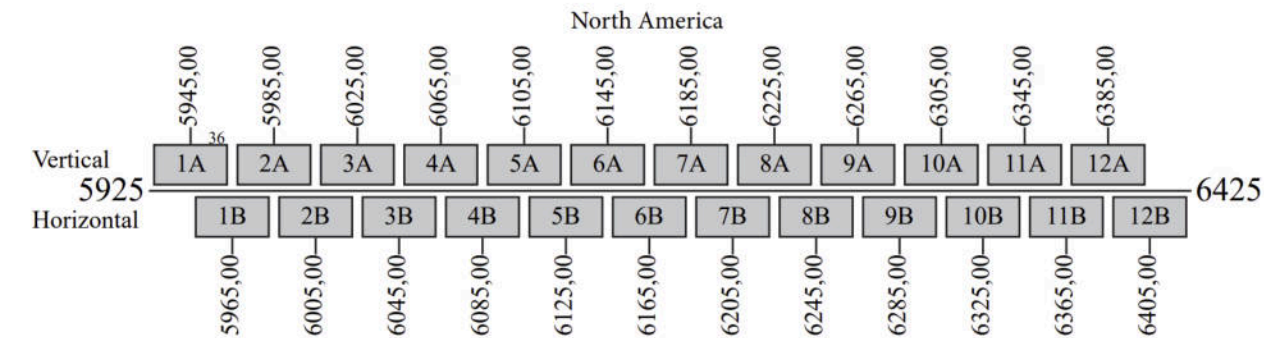
- **Dumb “bent pipes”**
 - Receives/bandpass filters uplink signals
 - Translates frequencies to downlink band
 - Linearly amplifies/bandpass filters downlink signal
- **Any RF signal received is retransmitted as-is!**
 - Not tied to any particular modulation scheme
 - No authentication either 🤖
- **Multiple uplinks can share a transponder**
 - Power levels must be approximately the same



Anik F1R Transponder Configuration

Uplink

Downlink



Our Uplink



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Our Uplink



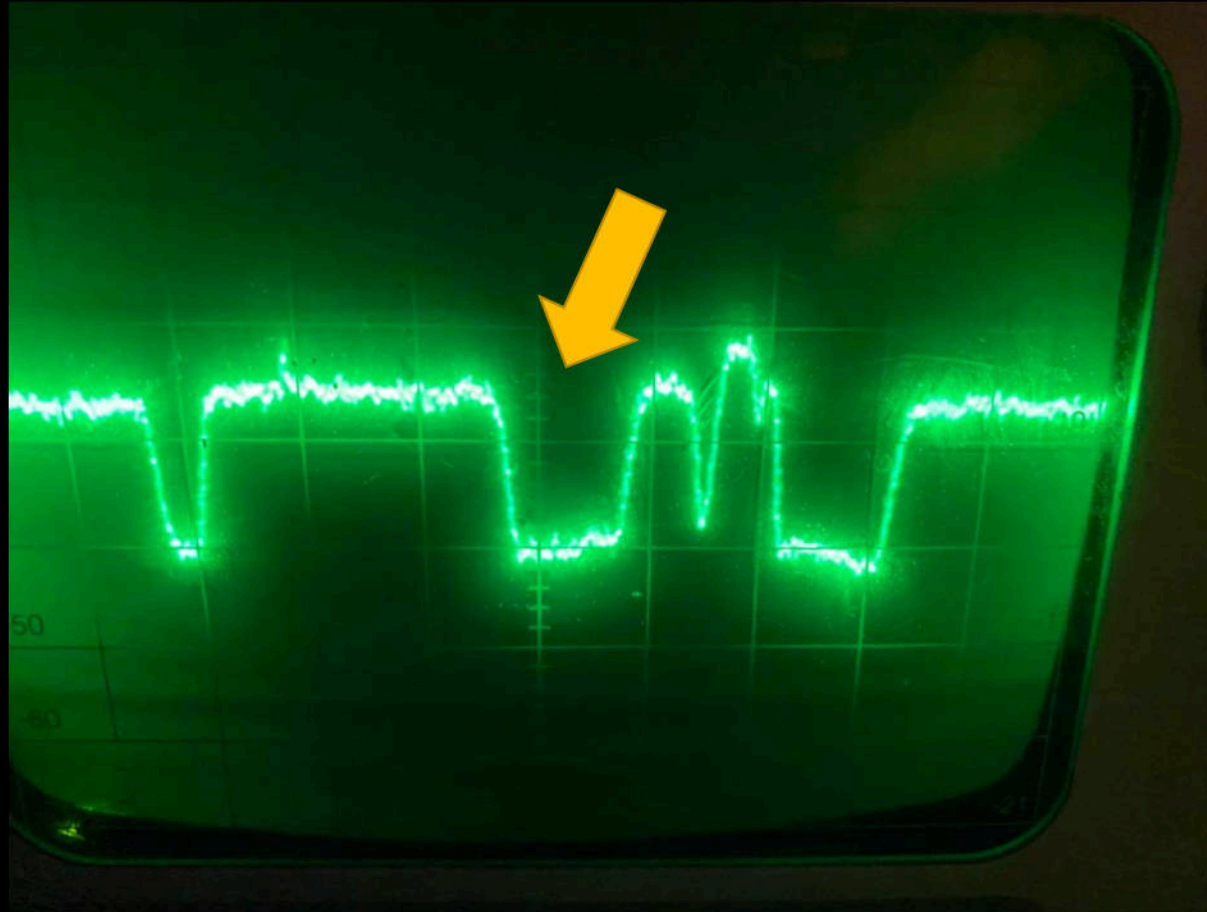
Our Uplink



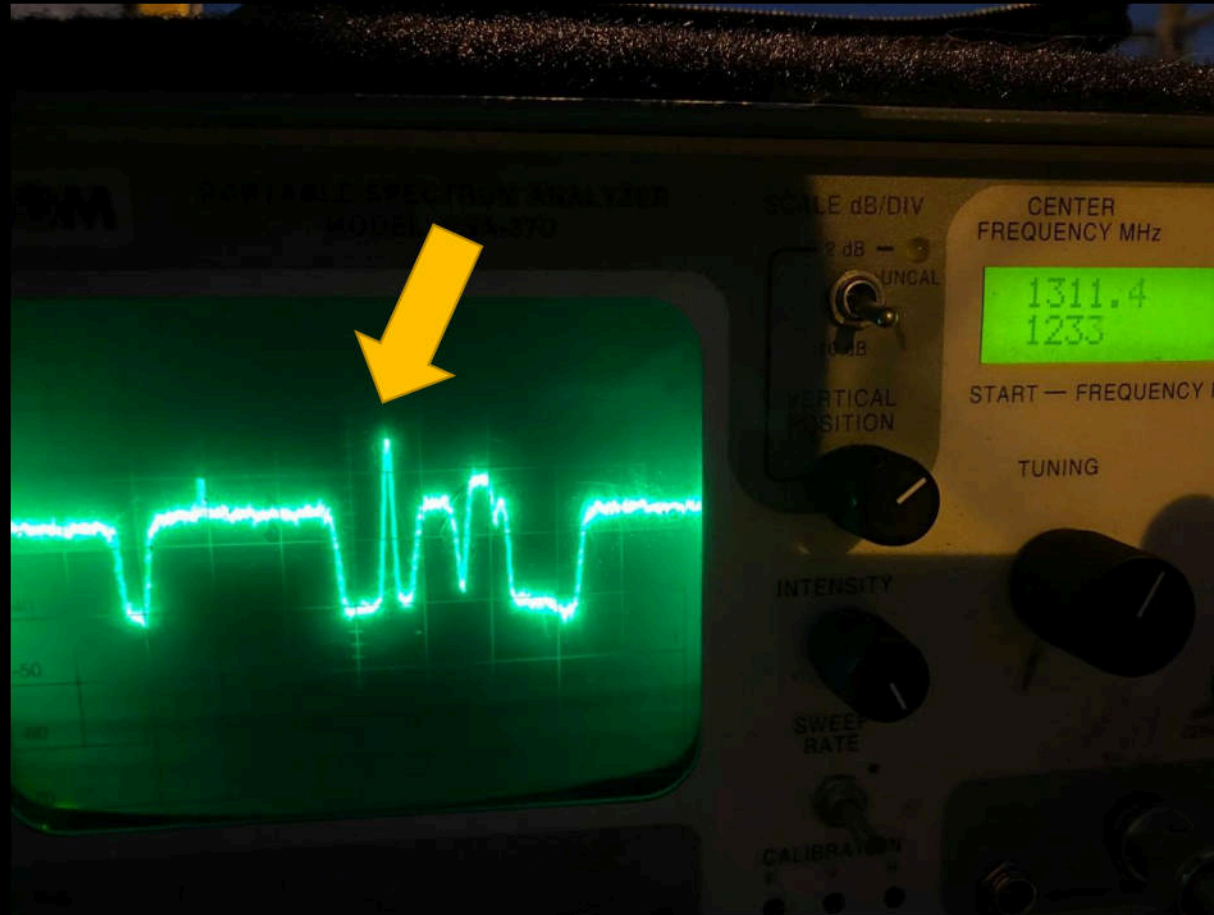
Our Uplink



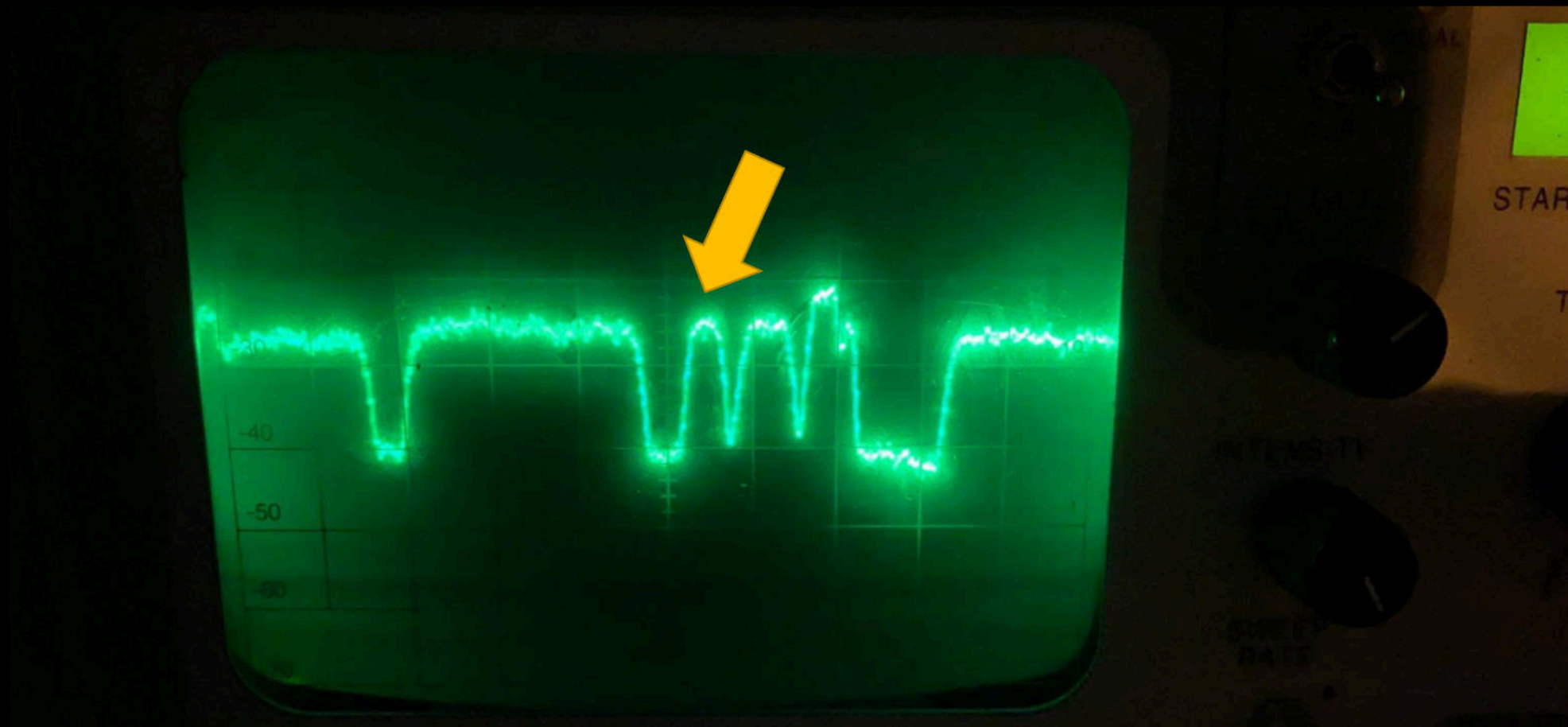
Testing the Uplink



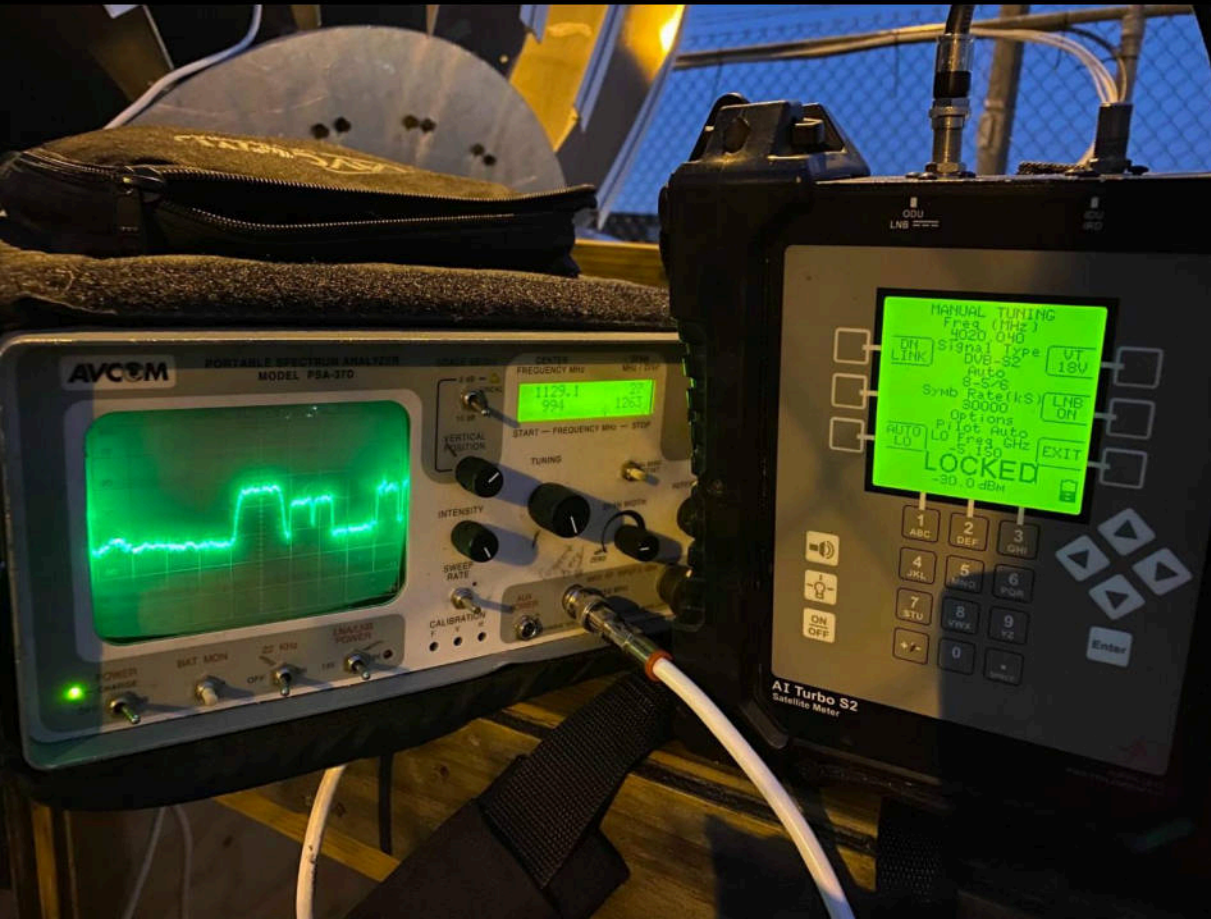
Testing the Uplink



Testing the Uplink



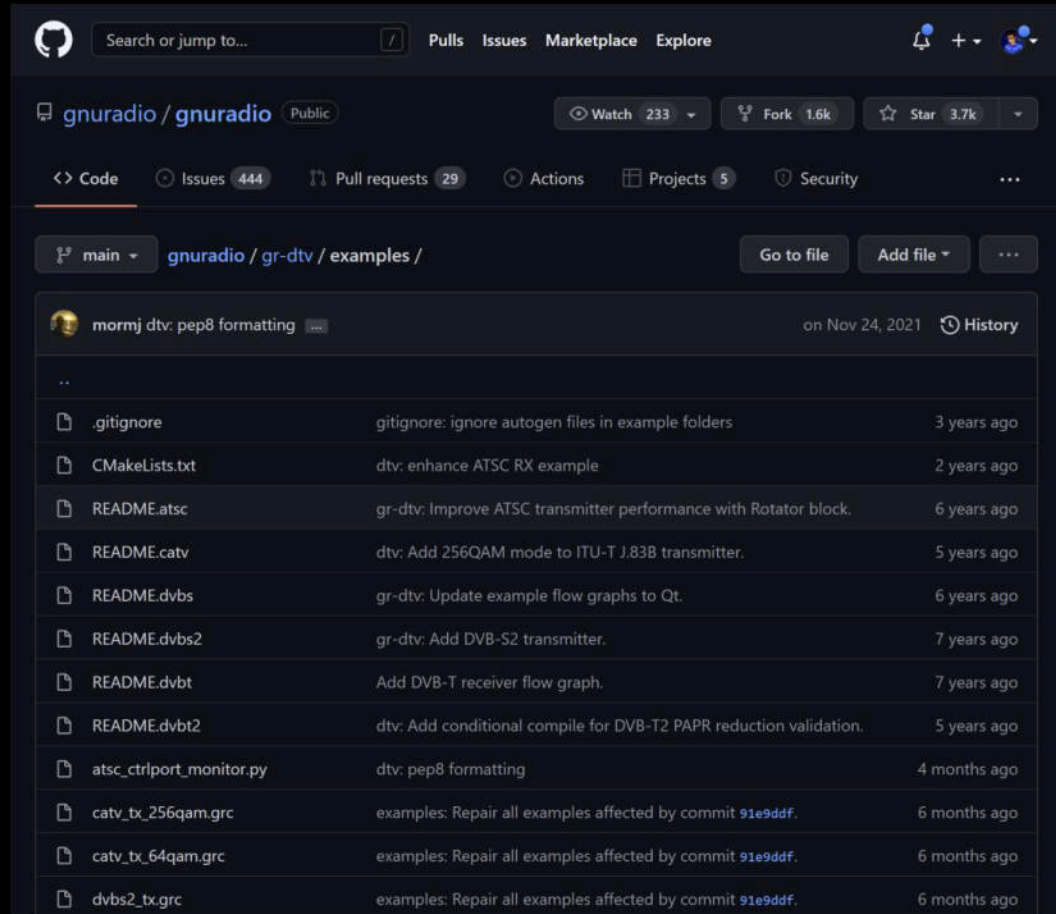
On-Air Testing



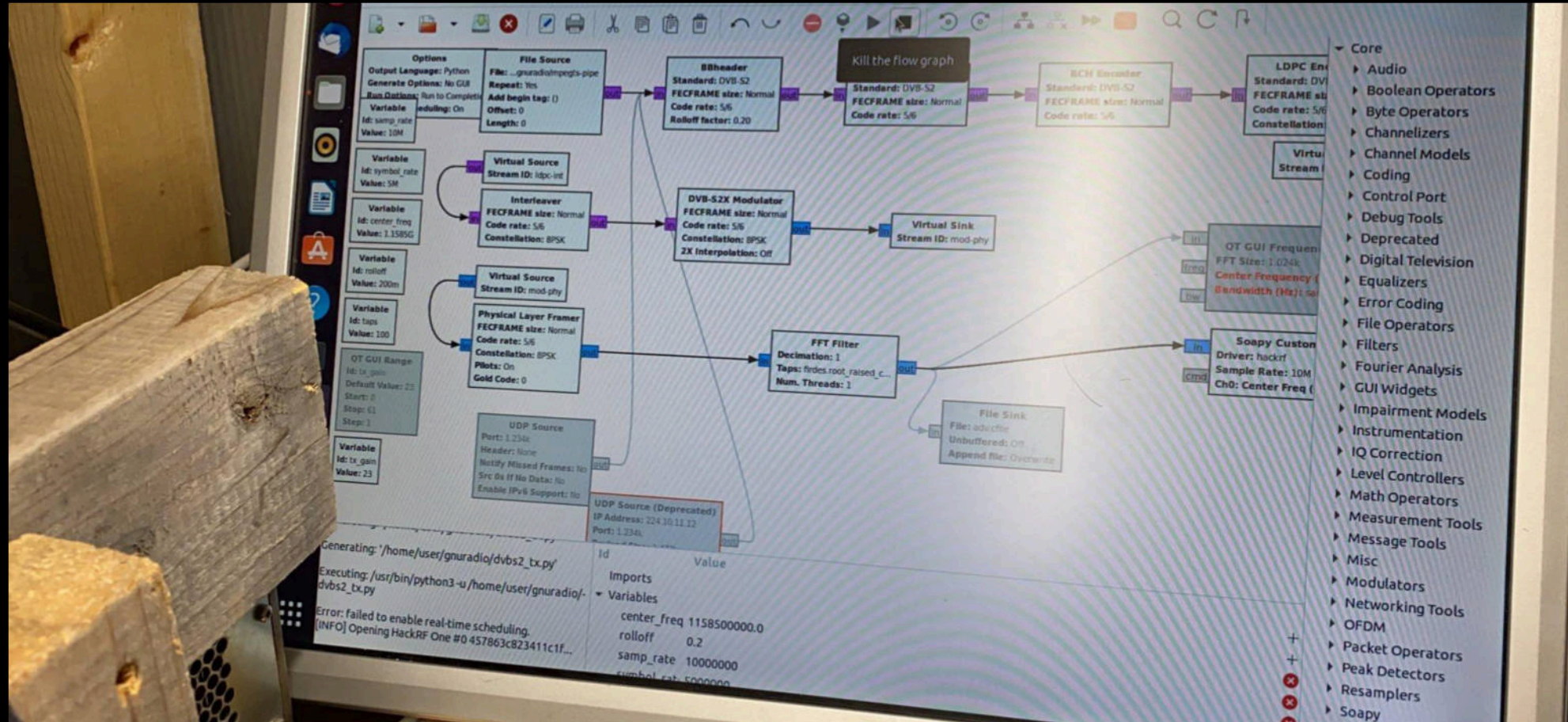
- **Tandberg Voyager E5740**
 - **Accepts ONLY ASI as input**



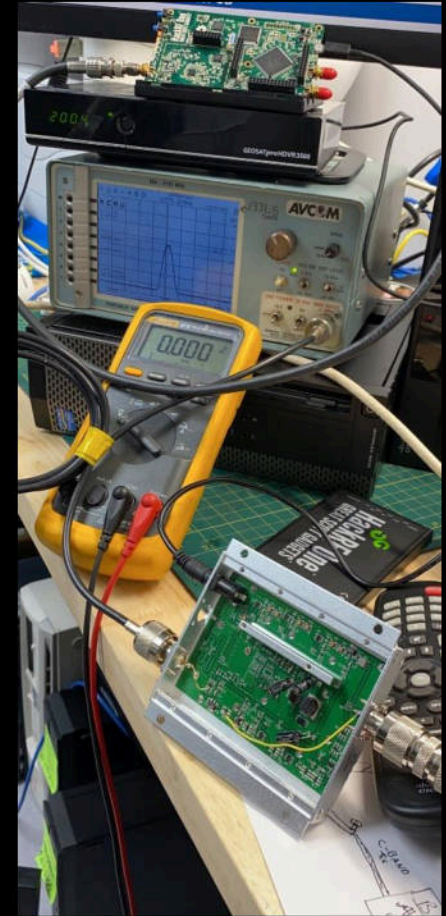
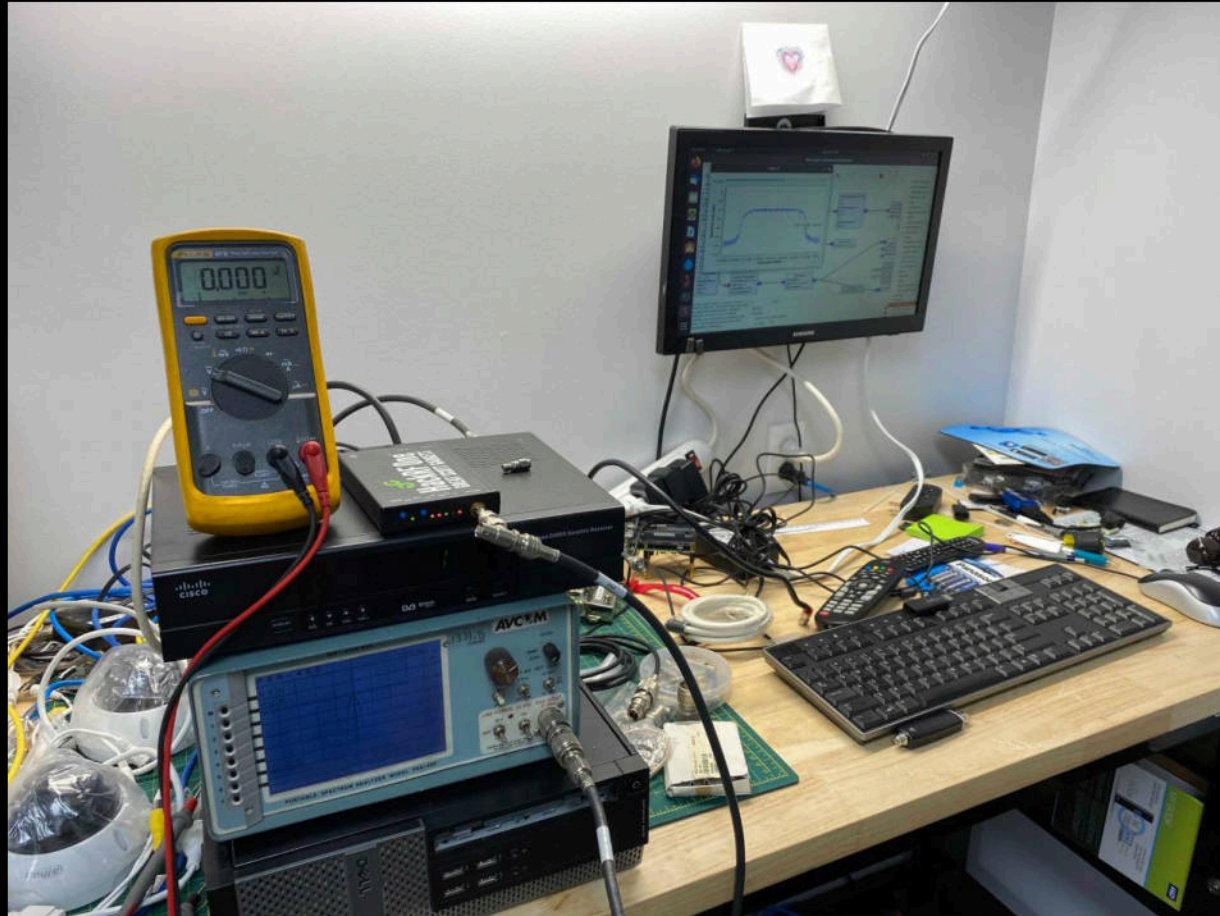
Building a DVB-S2 Transmitter



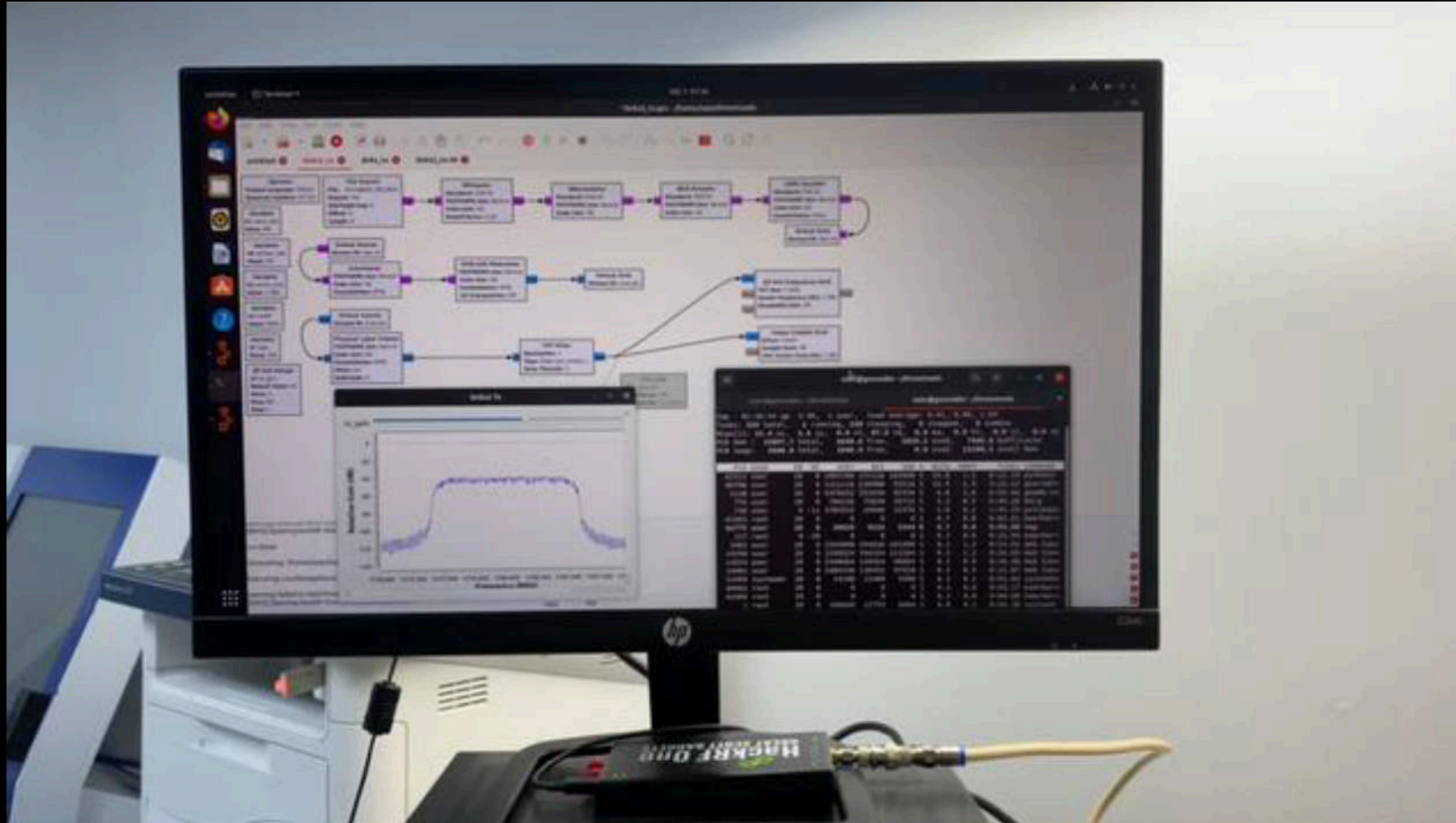
Building a DVB-S2 Transmitter



Building a DVB-S2 Transmitter



Building a DVB-S2 Transmitter



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Test stream: http://www.w6rz.net/sweetnothing8psk_184_56.ts

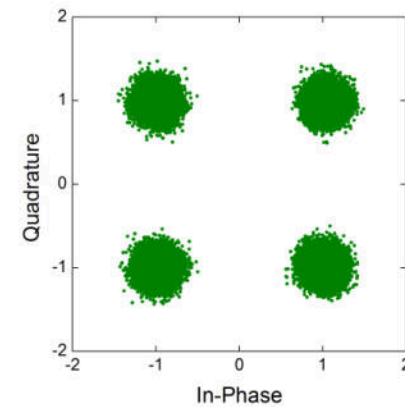
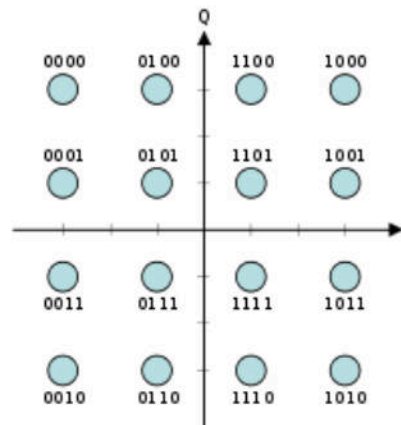
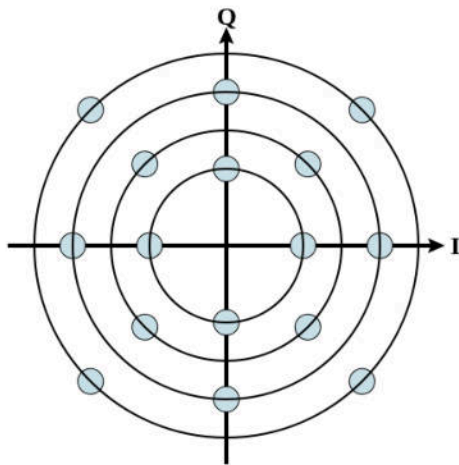
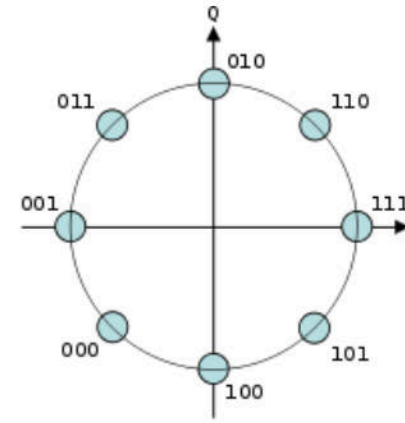
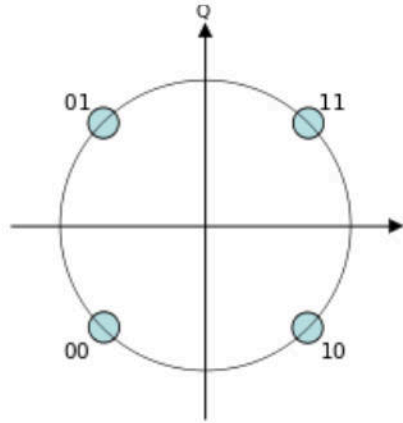
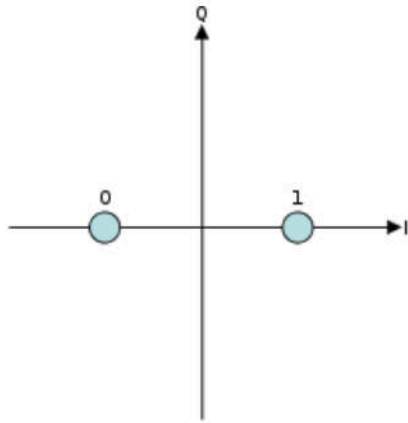
MPEG 2 Transport Streams

- **Carries one or more programs/channels over a continuous stream**
- **Used extensively: OTA TV, Cable TV, Satellites, even DOCSIS!**
- **Multiplexes media (ES) streams together, each assign its own PID**
- **Channel maps associate multiple ESes into a single program**
 - **e.g., one video stream, two audio streams in different languages**
 - **Can theoretically do multiple video streams, which some DVDs used**
 - **Codec-agnostic: can do H.262/H2.64/H.265/AC-3/MP3/AAC/etc.**

MPEG 2 Transport Streams

- **Broadcast M2TS requires extremely specific bit rates**
- **Derived from modulation parameters, e.g.,**
 - **Mode: (8VSB vs. PSK vs. OFDM)**
 - **Occupied bandwidth (e.g., 5-8 MHz)**
 - **Constellation / number of bits per symbol**
 - **Symbols per second**
 - **Error correction overhead (e.g., 1/2, 2/3, 3/4, 5/6, 7/8)**
 - Simple form: represent data as a degree (n-1) polynomial
 - Advanced forms can take advantage of “soft-decoding” (e.g., closeness to valid point)
- **Null packets (PID 0x1FFF) are inserted to keep bitrate constant**

Constellation Diagrams



MPEG 2 Transport Streams

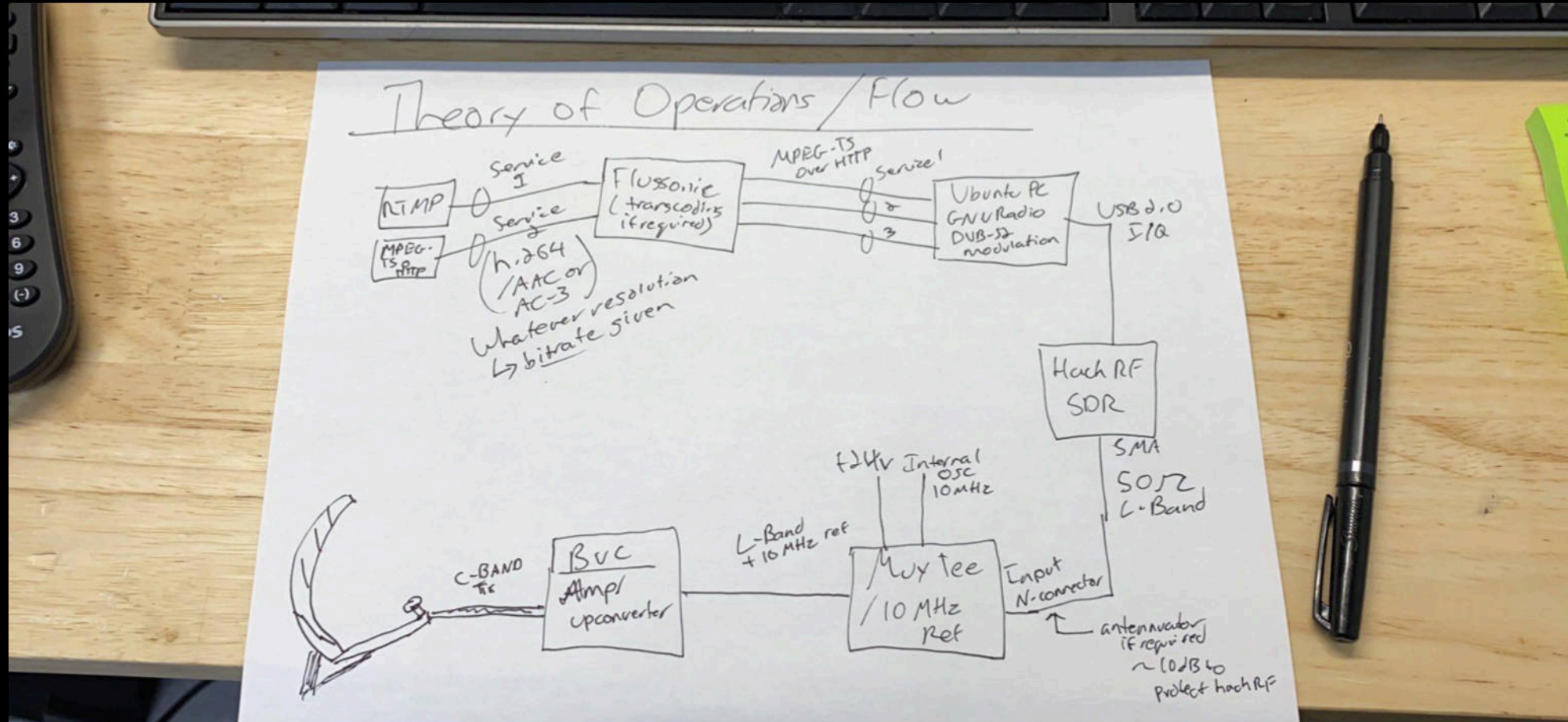
The screenshot displays the TSReader Lite application window, version 2.8.56. The interface is divided into several sections:

- Tree View (Left):** Shows the structure of the transport stream. The selected item is "PMT PID 0x0030 - Progr. 1". Other visible items include PAT PID 0x0000, VCT: W6RZ-DT, ES PID 0x0034, ES PID 0x0031, PCR PID 0x0031, MGT PID 0x1fff, TVCT PID 0x1fff, and 1 W6RZ-DT.
- General Information (Bottom Left):**
 - Source: Transport Stream File-loop
 - Tuner: sweetnothing8psk_184_56.ts
 - Signal: File Read: 13.4%
 - File Loops: 0
 - Network Type: ATSC
 - Run Time: 000:00:35
- MPEG-2 Statistics (Bottom Center):**

	PAT	PMT	CAT	ETT	PSIP	EIT
Sections	715	2	0	0	602	0
CRC Errors	0	0	0	0	0	0
Continuity Errors	0			Mux. bitrate:	4,469,943 bps	
TEI Errors	0			Last sec.:	4.428 Mbit	
				In buffer:		
				Out buffer:		
- Active PIDs (Center):** A list of active PIDs with their respective rates and bitrates.

PID	Rate	Bitrate
0x0031	74.96%	3.35 Mbps
0x1fff	14.01%	626.41 Kbps
0x0034	8.21%	411.79 Kbps
0x0030	0.67%	30.09 Kbps
0x0000	0.67%	30.09 Kbps
0x1fff	0.47%	21.09 Kbps
- Video Decode (Right):** A window showing the decoded video frame. The title is "37.1/1 - W6RZ-DT". The frame shows a night cityscape. Below the frame are buttons for "user", "cc", and "std".

The Complete Broadcast Chain



Media Chain

- **OBS generates RTMP source, pointed at our server (nginx)**
- **Flussonic ingests RTMP, performs PID remapping to support multiple programs, and produces M2TS over HTTP**
- **tsduck muxes multiple M2TSes together at final bitrate**
 - **Inserts null packets as needed**
- **Resulting stream sent to GNU Radio applet over socket**

Content Selection

- **What do we put in a hacker satellite broadcast?**
- **Why not an entire con?**
- **Satellite was going away in November 2021**
- **Got permission to restream Toorcon San Diego**
 - **Normally \$50!**



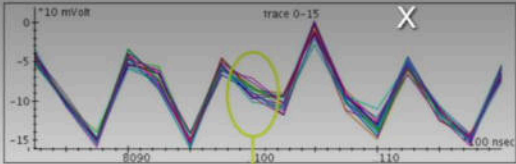
OBS Studio 27.1.1 (64-bit, windows) - Profile: Untitled - Scenes: Untitled

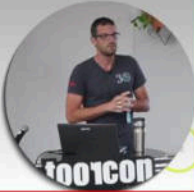
FileEditViewProfileScene CollectionToolsHelp

Preview

The media playback was aborted due to a corruption problem or because the media used features your browser did not support.







Transition

Quick Transitions

Cut

Fade (300ms)

Fade to Black (300ms)

Program



Toorcon Training

PropertiesFiltersInteractRefresh

Scenes

Shadytel 2

Antitrust Scene

Wargames Scene

Hackers

Hacker Moviez

Toorcon Scene

Sources

Toorcon Training

Audio Mixer

Desktop Audio

Hacker Movie List

Shadytel Logo

Scene Transitions

Fade

Duration 300 ms

Controls

Stop Streaming

Start Recording

Start Virtual Camera

Studio Mode

Settings

Exit

Dropped Frames 0 (0.0%)

LIVE: 26:13:27

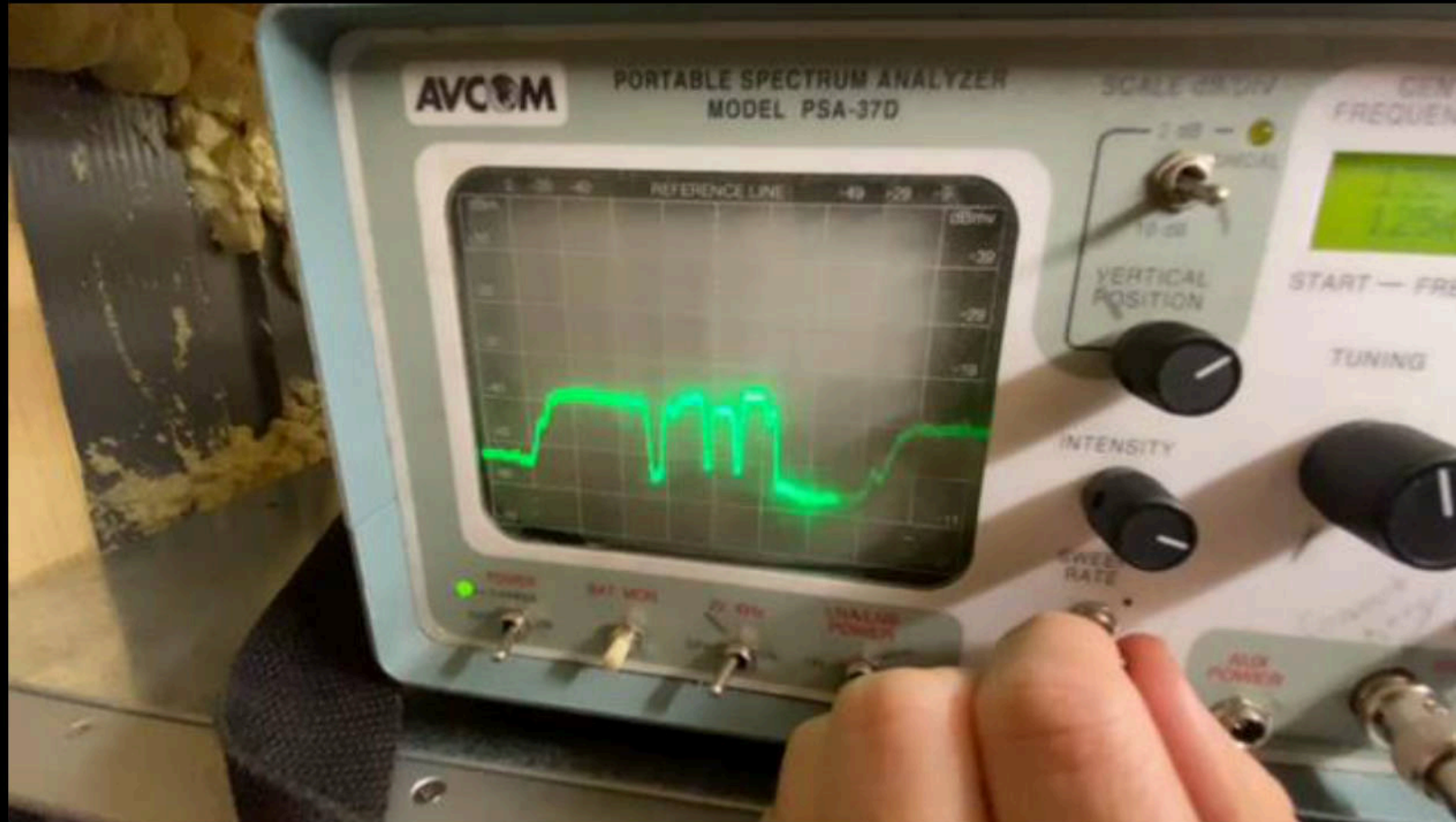
REC: 00:00:00

CPU: 72.6%, 29.97 fps

kb/s: 2095

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On-Air Testing



On-Air Testing



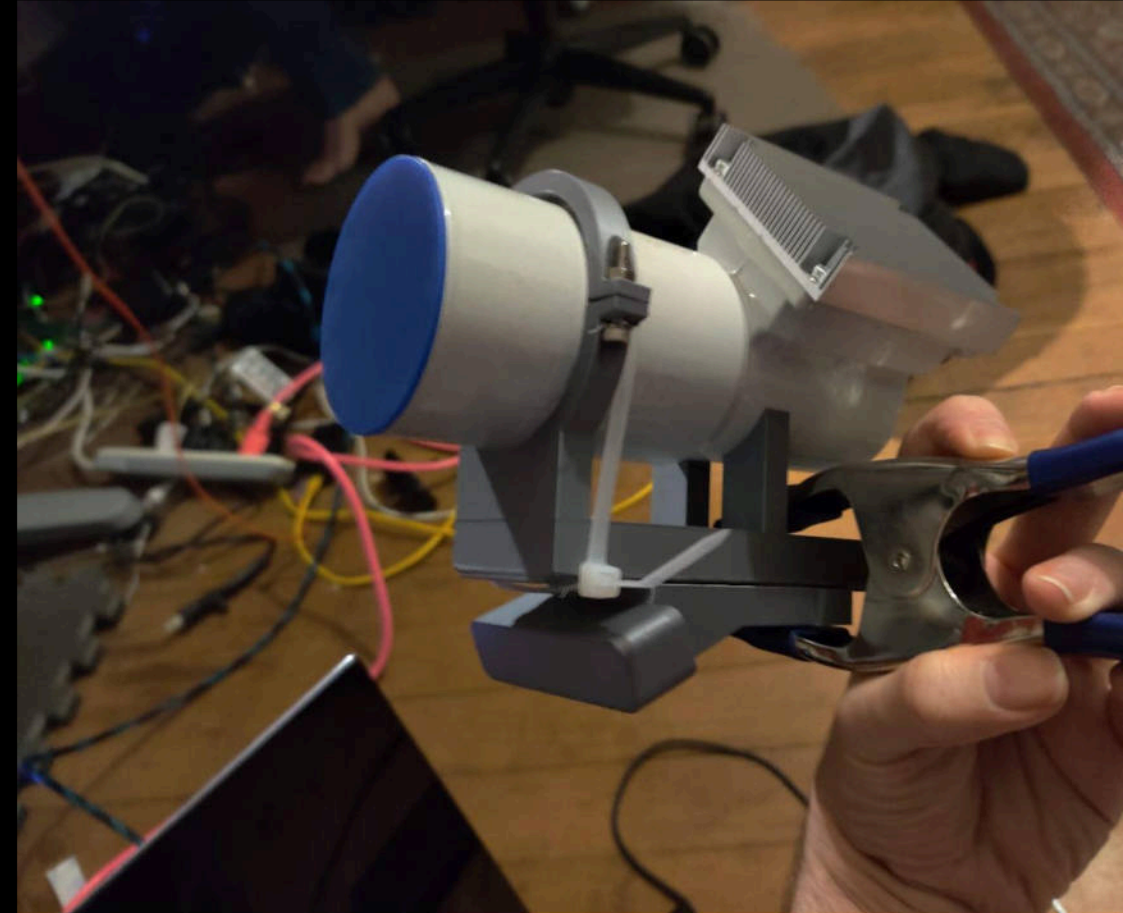
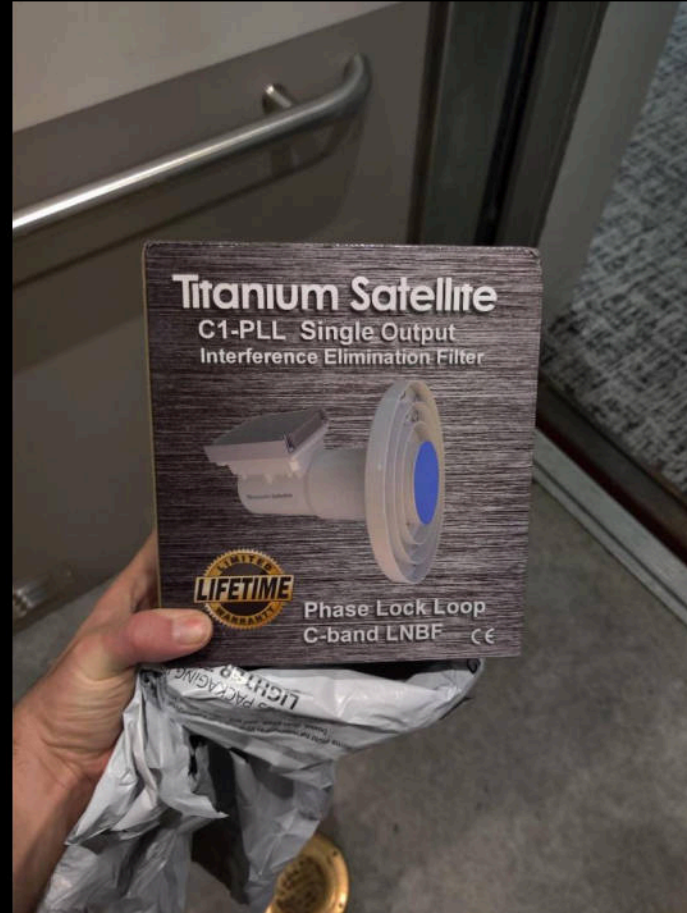
Receiving ShadyVision



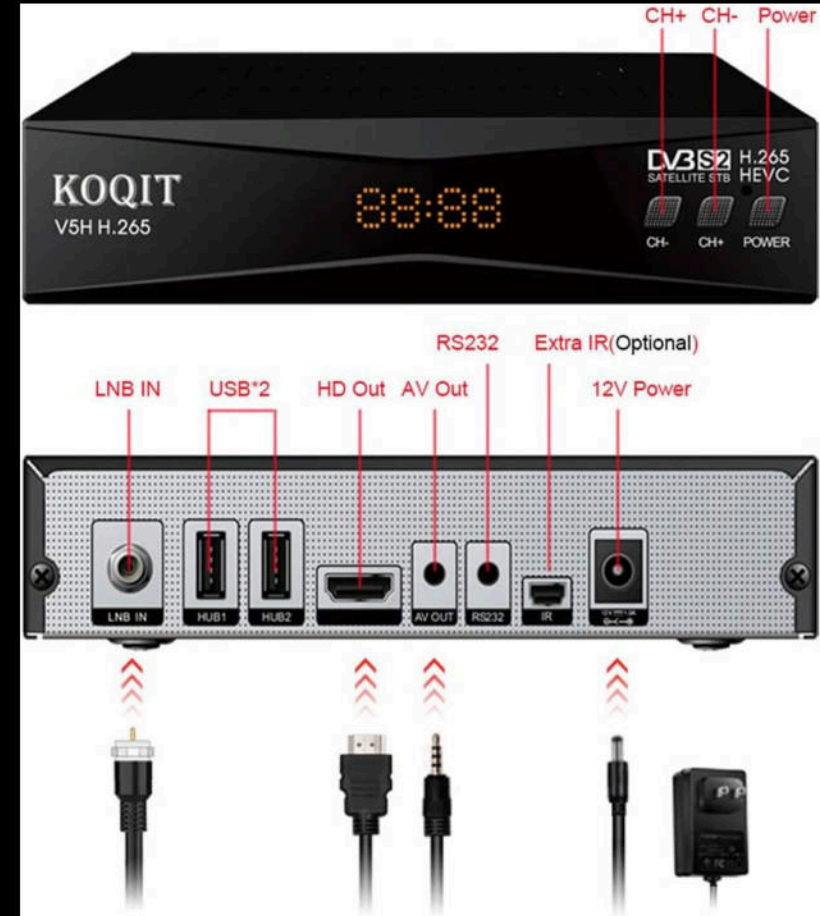
Receiving ShadyVision

- **CAUTION: Speculation below!**
- **About ~16 C/N at transmitter site (should be better in CONUS)**
- **Reusing a DirecTV dish would lose us about 14 dB**
- **... but remember, we control the modulation parameters!**
 - **DVB-S2, QPSK, 3/5 FEC requires 2.2 C/N**
 - **DVB-S2, QPSK, 1/2 FEC requires 1.0 C/N**
 - **DVB-S2, QPSK, 1/4 FEC requires -2.4 C/N**
 - **At 5000 kSym/s w/ pilot, effective throughput is 2388 kbps w/ ~4 dB link margin**
 - **DEF CON documentary at 1080p h265 *averages* 1595 kbps**

Receiving ShadyVision



Receiving ShadyVision



Receiving ShadyVision



Receiving ShadyVision



Receiving ShadyVision



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Receiving ShadyVision

Welcome to "Ricks Satellite"
In Association with Gofastmotorsports.com

Ku-Band Satellite Dishes
Shipping cost shown are for within the 48 cont. United States

Ku-Band Dishes	Picture & Part #	COST																												
DS2076 Winegard 76 CM Satellite Dish Made in the USA! Mounting Type: Ground, Pole & Wall Mount Adjustment Type: AZ/EL Comes with Pole and Pedestal Base Can be mounted to a Deck, Side of the House or on the Roof.	 DS2076 WINEGARD 76CM Satellite Dish Made in the USA!	DS2076 Winegard Satellite Dish - \$99.99 Including Shipping Within the United States \$ 74.99 Plus \$25.00 Shipping Within the 48 Cont. United States Add to Cart This dish does not work with motors. You should use the GeoSat Pro 90CM or 1.2 Meter if you want to use a Motor. This dish is large enough to still work for most channels available on Galaxy 19 @ 97 West. Most other N. American Satellites are using newer signals, like MPEG4 and MPEG5 with other modulations and you should use either a 90CM or 120CM Dish to have reliable reception.																												
GeoSatpro 90CM Offset Satellite Dish Technical Specifications <table><tbody><tr><td>Mechanical</td><td></td></tr><tr><td>Effective Aperture</td><td>90 cm / 36"</td></tr><tr><td>Diameter</td><td>93.5cm x 95 cm</td></tr><tr><td>Assembled Weight</td><td>12 lbs.</td></tr><tr><td>Reflector Thickness</td><td>.7mm</td></tr><tr><td>F/D</td><td>.5</td></tr><tr><td>Offset Angle</td><td>24.62 degrees</td></tr><tr><td>Surface Accuracy</td><td>±/- 0.01 deg."</td></tr><tr><td>Elevation Range</td><td>0 - 90 deg.</td></tr><tr><td>Finish</td><td>Textured Poly Powder Coat Color: Charcoal</td></tr><tr><td>Mount Type</td><td>Universal - Roof or Wall with adjustable position tripod legs, 27" 3" post double ended for height or reach setting. 360 degree level adjustment.</td></tr><tr><td>Feed Support</td><td>Universal 40mm / 23mm Clamp</td></tr><tr><td>Electrical</td><td></td></tr><tr><td>Frequency Range</td><td>10.95 - 12.75 GHz</td></tr></tbody></table>	Mechanical		Effective Aperture	90 cm / 36"	Diameter	93.5cm x 95 cm	Assembled Weight	12 lbs.	Reflector Thickness	.7mm	F/D	.5	Offset Angle	24.62 degrees	Surface Accuracy	±/- 0.01 deg."	Elevation Range	0 - 90 deg.	Finish	Textured Poly Powder Coat Color: Charcoal	Mount Type	Universal - Roof or Wall with adjustable position tripod legs, 27" 3" post double ended for height or reach setting. 360 degree level adjustment.	Feed Support	Universal 40mm / 23mm Clamp	Electrical		Frequency Range	10.95 - 12.75 GHz		GEOSATpro 90CM Offset Satellite Dish \$129.00 Includes Shipping Within the 48 Cont. United States Add to Cart Add a GEOSATpro SL1PLL Standard LNB for only \$10.00 More! Add to Cart The GEOSATpro 90cm has several exclusive features not found on any other satellite dish and provides the strongest mast clamp of any design. Six grips bite into the mast when tightened and prevents the the dish from slipping or rotating out of alignment. Three clamping bolt holes provide maximum clamping power for use in high wind regions. Four ribs evenly distribute the clamping pressure to prevent crushing the mast or post. Adjustable elevation bracket allows for dish aiming angles from 0 - 90 degrees. Hardware is hardened SAE with serrated locking washer nuts.
Mechanical																														
Effective Aperture	90 cm / 36"																													
Diameter	93.5cm x 95 cm																													
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Feed Support	Universal 40mm / 23mm Clamp																													
Electrical																														
Frequency Range	10.95 - 12.75 GHz																													

Receiving ShadyVision

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C-Band 120cm/135cm/150cm/180cm/210cm/240cm satellite dish antenna used outdoor

FOB Reference Price: [Get Latest Price](#)

\$99.00 - \$159.00 / piece | 1 piece/pieces (Min. Order)

Benefits: Quick refunds on orders under US \$1,000 [Claim now >](#)

quantity:   pieces

Customization:

Protection:  **Trade Assurance** Protects your Alibaba.com orders

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Anhui Bowei Electronics Technolog...
Manufacturer, Trading Company
 CN 14YR

Response Time: **≤2h** On-time delivery r...: **96.3%**

44 Transactions
20,000+

You may also like



C/Ku Band Dish Antenna for 1.8m 2.4m 3m

\$12.00 - \$150.00
1 piece (MOQ)



C Band 240cm (8 feet) Prime focus antenna...

\$99.00 - \$159.00
1 unit (MOQ)



Good Performance 240cm 2.4m 8Feet C Band Prime...

\$90.00 - \$150.00
1.0 piece (MOQ)



Mesh type satellite antenna C band antenna...

\$12.00 - \$150.00
1 piece (MOQ)



Galvanized Steel Board 3m 300cm 2.4m 240cm 1.8...

\$45.00 - \$550.00
1.0 piece (MOQ)

Setting Up Your Own Broadcasts

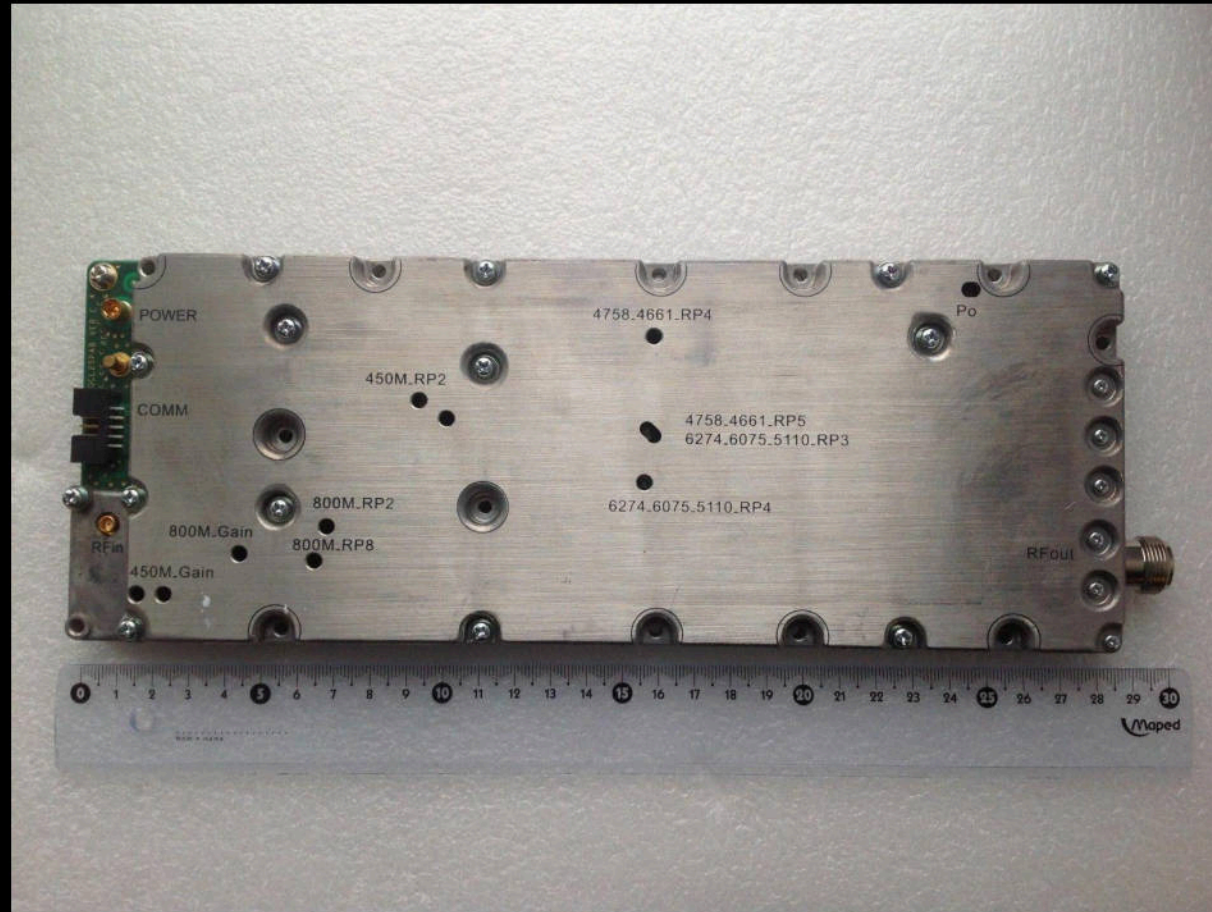
- **For C band satellite broadcasts, you need:**
 - A LARGE dish
 - A way to upconvert to ~6 GHz
 - A surprisingly small amplifier (we operated at ~15 Watts)
 - Permission 😊
- **Ku band operation requires more power and components that can run up to ~14.5 GHz**
 - Smaller wavelength means smaller dishes
 - Probably way more supply on eBay

Setting Up Your Own Broadcasts

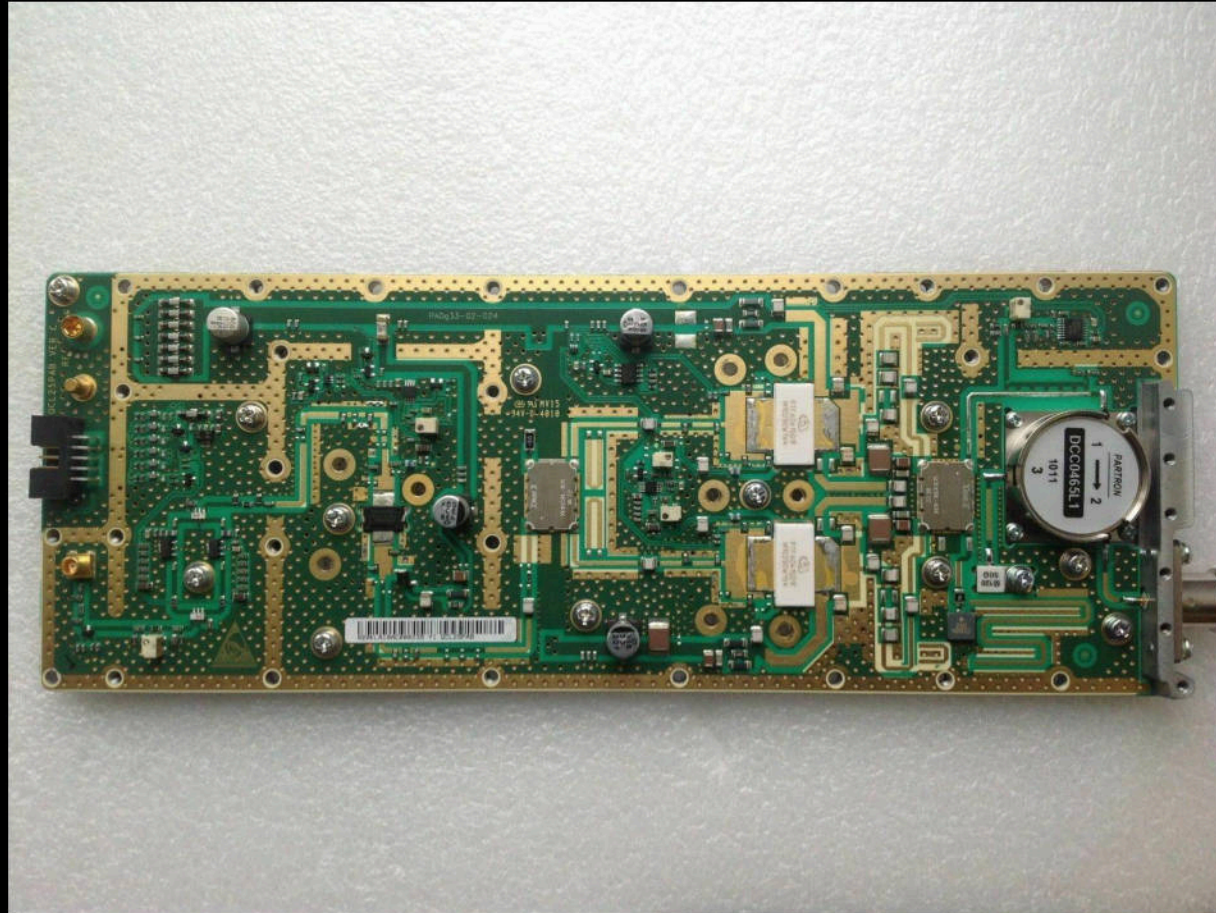
- **Almost nothing in this talk is specific to satellites!**
- **Use DVB-T for terrestrial broadcasts**
 - Everyone has a tuner – the RTL-SDR
 - Tweak modulation parameters (symbol rate, constellation, FEC) as desired to meet your link budget
 - Need to replace amplifier/antenna
 - Existing Amateur TV amplifiers? (Needs to be linear)
 - Sketchy Russian LTE amps??
 - Overlaps UHF band and 70cm Amateur spectrum
 - Please filter and monitor your emissions!



Setting Up Your Own Broadcasts



Setting Up Your Own Broadcasts



Setting Up Your Own Broadcasts

- ... and of course, permission 😊
- **One approach: constrain transmissions to "information bulletins" "directed only to amateur operators consisting solely of subject matter of direct interest to the amateur service." (i.e., this talk)**
- **Or: Obtain a Special Temporary Authority (STA) from the FCC for testing in un-used TV whitespace**
 - **Used by Shadytel to run a GSM network at Toorcamp one year**

Setting Up Your Own Broadcasts

United States of America FEDERAL COMMUNICATIONS COMMISSION EXPERIMENTAL SPECIAL TEMPORARY AUTHORIZATION				
EXPERIMENTAL (Nature of Service)		WF9XUL (Call Sign)		
XT FX (Class of Station)		0543-EX-ST-2012 (File Number)		
NAME Nicholas Burrows				
This Special Temporary Authorization is granted upon the express condition that it may be terminated by the Commission at any time without advance notice or hearing if in its discretion the need for such action arises. Nothing contained herein shall be construed as a finding by the Commission that the authority herein granted is or will be in the public interest beyond the express terms hereof.				
This Special Temporary Authorization shall not vest in the grantee any right to operate the station nor any right in the use of the frequencies designated in the authorization beyond the term hereof, nor in any other manner than authorized herein. Neither the authorization nor the right granted hereunder shall be assigned or otherwise transferred in violation of the Communications Act of 1934. This authorization is subject to the right of use of control the Government of the United States conferred by Section 706 of the Communications Act of 1934.				
Special Temporary Authority is hereby granted to operate the apparatus described below:				
Purpose Of Operation: Testing GSM network.				
Station Locations (1) Neah Bay (CLALLAM), WA - NL 48-20-13; WL 124-39-52				
Frequency Information Neah Bay (CLALLAM), WA - NL 48-20-13; WL 124-39-52				
Frequency 1958-1960 MHz	Station Class FX	Emission Designator 270KG7W	Authorized Power 1 W (ERP)	Frequency Tolerance (+/-)
Special Conditions: (1) All customers participating in the experiment shall be informed that this operation is for experimental purposes only and can be cancelled at any time. (2) Operation in the 1850-1910 or 1930-1990 MHz band requires prior frequency coordination with 2 GHz microwave users and consent of PCS licensees to avoid interference.				
This authorization effective August 07, 2012 and August 14, 2012 will expire 3:00 A.M. EST				
FEDERAL COMMUNICATIONS COMMISSION 				

Page 1 of 2

The purpose of the test will be to verify functionality of an Open Source GSM network design in an outdoor test range under load of up to 1,000 test users. For one week, users will send and receive text messages and place voice-calls to various services and other users while the system is under observation. System will consist of several Base Stations to provide experimental service to test users. Test users will be notified that the system is experimental, and access will be limited to SIM cards provided to test users. Test will occur for the duration and in the location of the Toorcamp Conference (08/06/2012-08/12/2012). The conference provides an ideal source of informed test users for this test.

Equipment is Commercial PicoCells (FCC ID: PEI-INSITE-1900), with the Test Users using standard quad-band GSM consumer devices. The PicoCells will be located below the surrounding tree-line and structures in the test site to help stress the system in a poor RF environment and limiting the system to line-of-sight operation.

What Is This Good For?!

- **With a large enough audience, *very* efficient use of bandwidth**
 - e.g., hacker events with limited WiFi
 - Range can be significantly larger
- **Works even if all network links are down**
- **A way to get information across borders?**
 - Works both ways: the EU can't block direct reception of RT
 - Caveat: the dishes are large and might also be restricted

The logo for 'shadytel' is centered on a dark gray rectangular background. The word 'shadytel' is written in a lowercase, sans-serif font with a vibrant green-to-blue gradient. Below it, the words 'THE PHONE COMPANY' are written in a smaller, all-caps, white sans-serif font.

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